

Kapitel 2 - Wahrscheinlichkeitsrechnung

$$\textcircled{2} P(X=k) = p^{k-1} (1-p)$$

geometrische Reiheformel:

$$\sum_{k=0}^{\infty} d^k = \frac{1-d}{1-d}$$

$$F_X(X) = \sum_{k=1}^{\infty} P(X=k) = \sum_{k=1}^{\infty} p^{k-1} (1-p) \quad \text{Substitution mit } m$$

$$F_X(X) = \sum_{m=0}^{k-1} p^m (1-p) = (1-p) \cdot \sum_{m=0}^{k-1} p^m = \frac{1-p^k}{1-p} \cdot (1-p) = 1-p^k \xrightarrow{\lim k \rightarrow \infty} 1$$

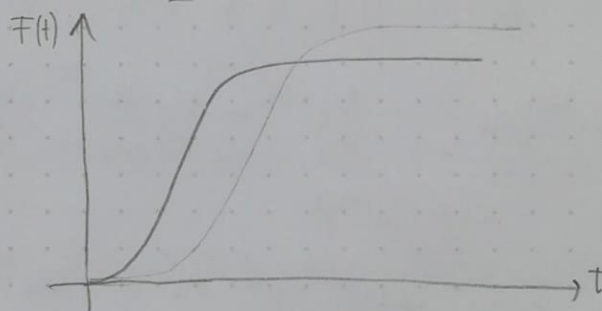
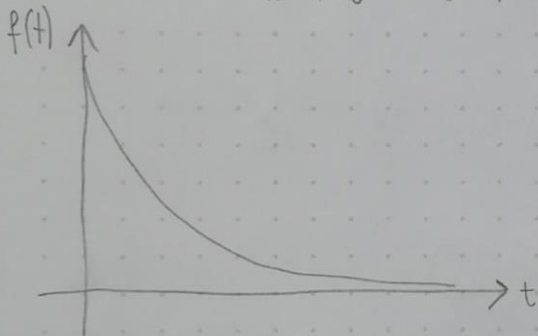
$$\textcircled{3} t (t \geq 0)$$

$$f(t) = \alpha \cdot e^{-\lambda t}$$

$$F(t) = \int_{-\infty}^{\infty} \alpha \cdot e^{-\lambda t} dt \stackrel{!}{=} 1 \quad (\text{Exponentialverteilung definiert von } 0 \leq x < \infty)$$

$$F(t) = \int_0^{\infty} \alpha \cdot e^{-\lambda t} dt \stackrel{!}{=} 1$$

$$1 = [\alpha \cdot e^{-\lambda t} \cdot (-\frac{1}{\lambda})]_0^{\infty} = -0 - (-\frac{\alpha}{\lambda}) \stackrel{!}{=} 1 \rightarrow \underline{\underline{\alpha = \lambda}}$$



$$\textcircled{4} F(t) = 1 - e^{-0.05 \cdot t}$$

$$a) P(T \geq 2a) = 1 - P(T \leq 2a) = 1 - (1 - e^{-0.05 \cdot 2}) = 90.5\%$$

$$b) P(T = 3a) = P(2a < T < 3a) = [1 - e^{-0.05 \cdot 3}] - [1 - e^{-0.05 \cdot 2}] = 0.0413 = 4.13\%$$

$$P(2a < T < 3a) = P(T \leq 3a) - P(T \leq 2a)$$

$$\int_{2a}^{3a} f(t) dt = \left[-\frac{1}{0.05} \cdot e^{-0.05t} \right]_{2a}^{3a} = \left[-e^{-0.05t} \right]_{2a}^{3a} = -e^{-0.05 \cdot 3} - (-e^{-0.05 \cdot 2}) = 4.13\%$$

$$c) p = \frac{1}{2}$$

$$\frac{1}{2} = 1 - e^{-0.05 \cdot t} \rightarrow -\frac{1}{2} = -e^{-0.05t} \rightarrow \ln\left(\frac{1}{2}\right) = -0.05t$$

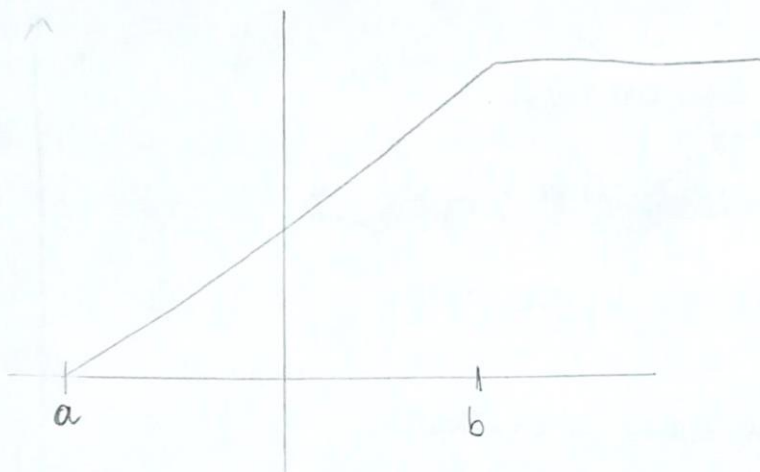
$$t = 13.86 \approx 13.9 \text{ Monate } 10 \text{ Tage}$$

⑤

$$f_X(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b \\ 0, & \text{ansonsten} \end{cases}$$

$$F_X(x) = \int_{-\infty}^{\infty} f_X(x) dx = \int_{-\infty}^a 0 dx + \int_a^x f_X(x) + \int_x^{\infty} 0 dx$$

$$= \int_{x=a}^{x=b} \frac{1}{b-a} dx = \left[\frac{x}{b-a} \right]_a^x = \frac{b}{b-a} - \frac{a}{b-a} = \frac{x-a}{b-a}$$



Warum Grenzen a bis x??

↳ ab b ist die Fkt nur noch 1

integriert werden kann, nachdem $F_X(x)$ von x abh. ist dennoch bis zu einem beliebigen x Wert

$$F_X(x) = \begin{cases} 0, & x \leq a \\ \frac{x-a}{b-a}, & a \leq x \leq b \\ 1, & x \geq b \end{cases}$$

$$F_X(x_1) = P(X \leq x_1)$$

⑥

Erwartungswert der Lebensdauerverteilung

$$E[X] = \int_{-\infty}^{\infty} \lambda e^{-\lambda t} dt \cdot t \quad \left[\frac{1}{\lambda} \quad t \right]$$

$$E[X] = \int_{-\infty}^0 0 dt + \int_0^{\infty} t \lambda e^{-\lambda t} dt \quad t=u \quad u'=1 \quad v'=\lambda e^{-\lambda t} \quad v=-\frac{\lambda}{\lambda} e^{-\lambda t}$$

$$\left[-\frac{t}{\lambda} e^{-\lambda t} - \int_0^{\infty} (-e^{-\lambda t}) dt \right] = \left[-\frac{t}{\lambda} e^{-\lambda t} - \frac{1}{\lambda} e^{-\lambda t} \right]_0^{\infty} = \left[(0-0) - \left(-\frac{1}{\lambda} \right) \right] = \underline{\underline{\frac{1}{\lambda}}}$$

$$\textcircled{7} \quad \text{Var}[X] = E[(X-\mu)^2] \quad E[X] = \frac{1}{\lambda} = \mu$$

$$E[(X^2 - 2X\mu + \mu^2)] = E[X^2] - 2(E[X])^2 + \mu^2 = m_2 - \mu^2 = m_2 - \left(\frac{1}{\lambda^2}\right) \quad \textcircled{I}$$

$$m_2 = \int_0^{\infty} \lambda t^2 e^{-\lambda t} dt \quad u=t^2 \quad u'=2t \quad v'=\lambda e^{-\lambda t} \quad v=-e^{-\lambda t}$$

$$-t^2 \cdot e^{-\lambda t} - \int 2t e^{-\lambda t} dt \quad -2t=u \quad -2=u' \quad e^{-\lambda t}=v' \quad e^{-\lambda t} \cdot \left(-\frac{1}{\lambda}\right) = v$$

$$-t^2 \cdot e^{-\lambda t} - \left[\left(2t \cdot e^{-\lambda t} \cdot \frac{1}{\lambda} \right) - \int \frac{2}{\lambda} e^{-\lambda t} dt \right]$$

$$= \left[-t^2 e^{-\lambda t} - \left[\left(2t \cdot e^{-\lambda t} \cdot \frac{1}{\lambda} \right) - \left[-\frac{2}{\lambda^2} e^{-\lambda t} \right] \right] \right]_0^{\infty}$$

$$0 - \left(-\frac{2}{\lambda^2} \right) = \underline{\underline{\frac{2}{\lambda^2}}} \quad \text{einsetzen} \quad \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \underline{\underline{\frac{1}{\lambda^2}}}$$

⑧ 2.31:

$$f_X(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b \\ 0, & \text{ansonsten} \end{cases}$$

$$E[X] = \int_{-\infty}^{\infty} \frac{x}{b-a} dx = \int_{-\infty}^a 0 dx + \int_a^b \frac{x}{b-a} dx + \int_b^{\infty} 0 dx$$

$$E[X] = \int_{x=a}^{x=b} \frac{x}{b-a} dx = \left[\frac{x^2}{2(b-a)} \right]_a^b = \left[\frac{(b^2 - a^2)}{2(b-a)} \right] = \frac{b+a}{2}$$

$$m_2 = \int_{-\infty}^b 0 dx + \int_a^b \frac{x^2}{b-a} dx + \int_b^{\infty} 0 dx$$

$$m_2 = \left[\frac{x^3}{3(b-a)} \right]_a^b = \frac{b^3 - a^3}{3(b-a)} = \frac{(b-a) \cdot (b^2 + ba + a^2)}{3(b-a)} = \frac{1}{3}(b^2 + ba + a^2)$$

$$\begin{aligned} \text{Var}[X] &= m_2 - \mu^2 = \frac{1}{3}(b^2 + ab + a^2) - \left(\frac{a+b}{2}\right)^2 \\ &= \frac{1}{3} \frac{b^2 + ab + a^2}{1} - \frac{1}{4} \frac{a^2 + 2ab + b^2}{1} = \frac{4b^2 - 3b^2 + 4a^2 - 3a^2 + 4ab - 6ab}{12} = \frac{a^2 - 2ab + b^2}{12} = \frac{(a-b)^2}{12} \end{aligned}$$

$$b^3 - a^3 = (b-a) \cdot (a^2 + ab + b^2) \rightarrow \text{abkürzt} \rightarrow \text{Rechenweg??}$$

$$\textcircled{9} X(0,1) \Rightarrow \mu_X = 0 \quad \sigma_X = 1$$

$$E[Z] = E\left[\frac{X - \mu_X}{\sigma_X}\right] = \frac{E[X] - E[X]}{\sigma_X} = 0$$

$$\begin{aligned} \text{Var}[Z] &= \text{Var}\left[\frac{X - \mu_X}{\sigma_X}\right] = \text{Var}\left[\frac{X}{\sigma_X}\right] - \text{Var}\left[\frac{\mu_X}{\sigma_X}\right] \\ &= \text{Var}[X] \cdot \left(\frac{1}{\sigma_X}\right)^2 = \frac{\sigma_X^2}{\sigma_X^2} = 1 \end{aligned}$$

$$\textcircled{10} a) \begin{array}{cc} X & 1 & 2 & 3 & 4 & 5 \\ p_X & 0,5 & 0,3 & 0,1 & 0,05 & 0,05 \end{array}$$

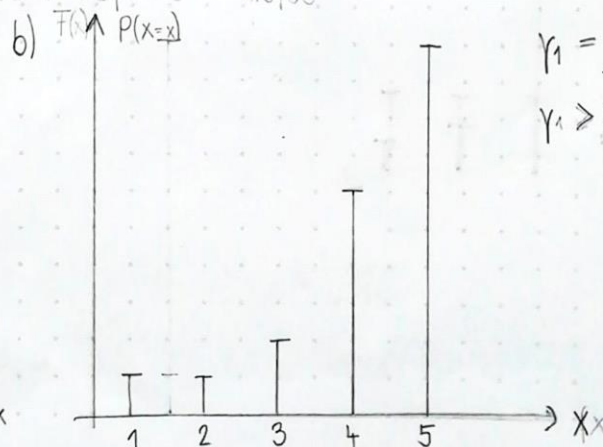
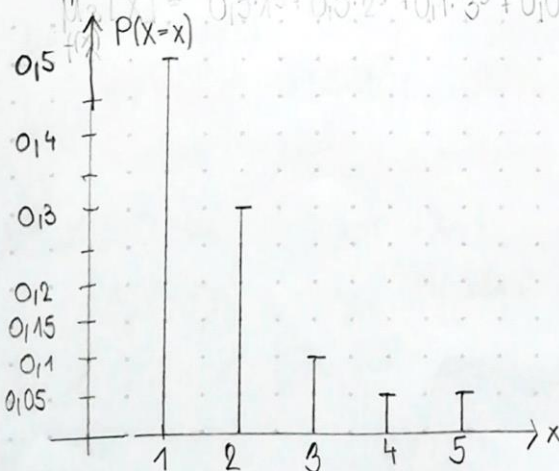
$$\mu = \sum_{i=1}^5 X_i \cdot f(X_i) = 0,5 \cdot 1 + 0,3 \cdot 2 + 0,1 \cdot 3 + 0,05 \cdot 4 + 0,05 \cdot 5 = 1,85 \rightarrow \mu^2 = 3,4225$$

$$E[X^2] = m_2 = \sum_{i=1}^5 X_i^2 \cdot f(X_i) = 0,5 \cdot 1^2 + 0,3 \cdot 2^2 + 0,1 \cdot 3^2 + 0,05 \cdot 4^2 + 0,05 \cdot 5^2 = 4,65 = 48,5\%$$

$$\text{Var}[X] = m_2 - \mu^2 = 4,65 - 3,4225 = 1,2275 \quad E[X^3] = m_3 = \sum_{i=1}^5 X_i^3 \cdot f(X_i) = 15,05$$

$$\gamma_1[X] = \frac{\mu_3[X]}{\sigma[X]^3} = \frac{E[(X - \mu)^3]}{\sigma[X]^3} = \frac{E[X^3 - 3X^2\mu + 3X\mu^2 - \mu^3]}{\sigma[X]^3} = \frac{E[X^3] - 3E[X^2] \cdot \mu + 3E[X] \cdot (\mu)^2 - \mu^3}{\sigma[X]^3}$$

$$\gamma_1[X] = \frac{(E[X^3] - 3\mu \cdot m_2 + 3\mu^2 \cdot \mu - \mu^3)}{\sigma[X]^3} = \frac{m_3 - 3 \cdot m_2 \cdot \mu + 2\mu^3}{\sigma[X]^3} = \frac{15,05 - 3 \cdot 4,65 \cdot 1,85 + 2 \cdot 1,85^3}{1,2275^3}$$



$$\gamma_1 = 1,1404$$

$\gamma_1 > 0 \rightarrow \text{rechtsschief}$

$$b) E[X] = \mu = \sum_{i=1}^5 x_i \cdot p_{xi} = 0,05 \cdot 1 + 0,05 \cdot 2 + 0,1 \cdot 3 + 0,3 \cdot 4 + 0,5 \cdot 5 = 4,15$$

$$m_2 = \sum_{i=1}^5 x_i^2 \cdot p_{xi} = 0,05 \cdot 1 + 0,05 \cdot 2^2 + 0,1 \cdot 3^2 + 0,3 \cdot 4^2 + 0,5 \cdot 5^2 = 18,15$$

$$\text{Var}[X] = m_2 - \mu^2 = 18,15 - 17,2225 = 0,9275$$

$$m_3 = \sum_{i=1}^5 x_i^3 \cdot p_{xi} = 0,05 \cdot 1 + 0,05 \cdot 2^3 + 0,1 \cdot 3^3 + 0,3 \cdot 4^3 + 0,5 \cdot 5^3 = 84,85$$

$$\gamma_1 = \frac{m_3 - 3 \cdot m_2 \cdot \mu + 2 \cdot \mu^3}{\sigma^3} = \frac{E[(X-\mu)^3]}{\sigma^3} = \frac{E[X^3 - 3X^2\mu + 3X\mu^2 - \mu^3]}{\sigma^3} = \frac{E[X^3] - 3E[X^2] \cdot \mu + 3E[X] \cdot (E[X])^2 - (E[X])^3}{\sigma^3}$$

$$\gamma_1 = \frac{m_3 - 3 \cdot m_2 \cdot \mu + 2 \cdot \mu^3}{\sigma^3} = \frac{84,85 - 3 \cdot 18,15 \cdot 4,15 + 2 \cdot 4,15^3}{\sqrt{0,9275}^3} = -1,401 \Rightarrow \gamma_1 < 0 \rightarrow \text{linksschief}$$

$$(12) a) \gamma_2 = \frac{m_4 - 4 \cdot m_3 \cdot \mu + 6 \cdot m_2 \cdot \mu^2 - 3 \cdot \mu^4}{\sigma^4} = \frac{E[(X-\mu)^4]}{\sigma^4} = \frac{E[X^4 - 4X^3\mu + 6X^2\mu^2 - 4X\mu^3 + \mu^4]}{\sigma^4} = \frac{m_4 - 4m_3\mu + 6m_2\mu^2 - 4\mu^4 + \mu^4}{\sigma^4}$$

$$\gamma_2 = \frac{m_4 - 4 \cdot m_3 \cdot \mu + 6 \cdot m_2 \cdot \mu^2 - 3 \cdot \mu^4}{\sigma^4} - 3$$

$$m_1 = \mu = \sum_{i=1}^5 x_i \cdot f(x_i) = 0,05 \cdot 1 + 0,1 \cdot 2 + 0,7 \cdot 3 + 0,1 \cdot 4 + 0,05 \cdot 5 = 3$$

$$m_2 = \sum_{i=1}^5 x_i^2 \cdot f(x_i) = 9,6$$

$$\text{Var}[X] = m_2 - \mu^2 = 9,6 - 3^2 = 0,6 = \sigma^2$$

$$m_3 = \sum_{i=1}^5 x_i^3 \cdot f(x_i) = 32,4$$

$$\gamma_1 = \frac{32,4 - 3 \cdot 9,6 + 2 \cdot 3^3}{\sigma^3} = 0$$

$$m_4 = \sum_{i=1}^5 x_i^4 \cdot f(x_i) = 115,2$$

$$\gamma_2 = \frac{115,2 - 4 \cdot 32,4 \cdot 3 + 6 \cdot 9,6 \cdot 3^2 - 3 \cdot 3^4}{\sigma^4} - 3 = -\frac{1}{3} = -0,3$$

$$b) m_1 = 3$$

$$m_3 = 45$$

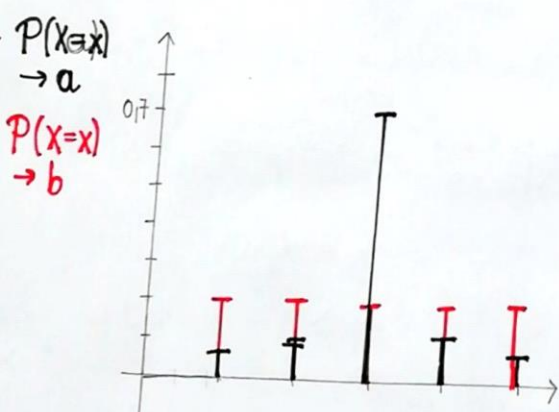
$$\text{Var}[X] = m_2 - \mu^2 = 11 - 3^2 = 2$$

$$m_2 = 11$$

$$m_4 = 195,8$$

$$\gamma_2 = \frac{195,8 - 4 \cdot 45 \cdot 3 + 6 \cdot 11 \cdot 9 - 3 \cdot 3^4}{\sigma^4} - 3 = -1,3$$

$$\gamma_1 = \frac{45 - 3 \cdot 11 \cdot 3 + 2 \cdot 3^3}{\sqrt{2}^3} = 0$$



$$(13) X \sim N(\mu, \sigma^2)$$

$$P(\beta > 150) = 1 - \Phi\left(\frac{\beta - \mu}{\sigma}\right) = 1 - \Phi\left(\frac{150 - 100}{40}\right) = 1 - \Phi(1,25) = 1 - 89,44\% = 10,56\%$$

$$(14) X \sim N(\mu, \sigma^2)$$

$$P(\mu - \sigma < X < \mu + \sigma): z = \frac{x - \mu}{\sigma} \rightarrow \frac{\mu \pm \sigma - \mu}{\sigma} = \pm 1$$

$$\Phi(1) = 84,13\%$$

$$\Phi(-1) = 15,87\%$$

$$P(\mu - \sigma < X < \mu + \sigma) = \Phi(1) - \Phi(-1) = 84,13\% - 15,87\% = \underline{68,26\%}$$

$$P(\mu - 2\sigma < X < \mu + 2\sigma): z = \frac{\mu \pm 2\sigma - \mu}{\sigma} = \pm 2$$

$$\Phi(2) = 97,72\%$$

$$1 - \Phi(2) = 2,28\%$$

$$P(\mu - 2\sigma < X < \mu + 2\sigma) = \Phi(2) - \Phi(-2) = 97,72\% - 2,28\% = \underline{95,44\%}$$

$$P(\mu - 3\sigma < X < \mu + 3\sigma): z = \pm 3$$

$$\Phi(3) = 99,865\%$$

$$\Phi(-3) = 0,135\%$$

$$P(\mu - 3\sigma < X < \mu + 3\sigma) = \Phi(3) - \Phi(-3) = \underline{99,74\%}$$

$$(15) Q_{1/2} = \mu$$

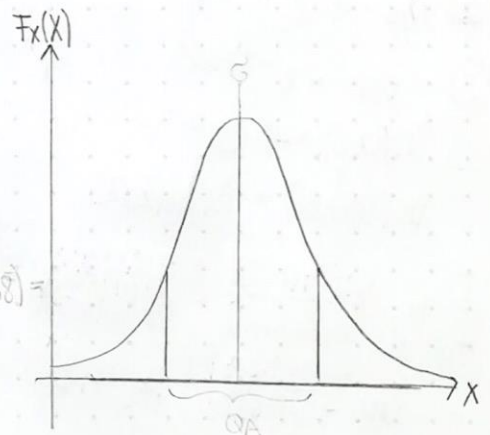
$$\frac{1}{2} = \Phi\left(\frac{x_{1/2} - \mu}{\sigma}\right) \Rightarrow 0 \text{ d.h. } \frac{x_{1/2} - \mu}{\sigma} = 0 \rightarrow x_{1/2} = \mu$$

$$Q_{3/4} - Q_{1/4} = \Phi\left(\frac{x_{3/4} - \mu}{\sigma}\right) - \Phi\left(\frac{x_{1/4} - \mu}{\sigma}\right) =$$

$$\Phi\left(\frac{x_{3/4} - \mu}{\sigma}\right) = \frac{3}{4} \Rightarrow \Phi^{-1}\left(\frac{3}{4}\right) = 0,68 \rightarrow x_{3/4} = 0,68 \cdot \sigma + \mu$$

$$\Phi\left(\frac{x_{1/4} - \mu}{\sigma}\right) = \frac{1}{4} \Rightarrow \Phi^{-1}\left(\frac{1}{4}\right) = -0,68 \rightarrow x_{1/4} = -0,68 \cdot \sigma + \mu$$

$$QA = 0,68 \cdot \sigma + \mu + 0,68 \cdot \sigma + \mu = 1,36 \cdot \sigma + 2\mu \quad (\mu = 0) \\ \rightarrow 1,36 \cdot \sigma$$



$$(16) Q_{1/4} - 1,5 \cdot QA = -0,68\sigma - 1,5 \cdot 1,36\sigma = -2,69\sigma$$

$$Q_{3/4} + 1,5 \cdot QA = 0,68\sigma + 1,5 \cdot 1,36\sigma = 2,69\sigma$$

$$\Phi(+2,69) - (-2,69) = 0,99643 + (1 - 0,99643) = 99,286 \rightarrow 1 -$$

$$\text{Auswertung: } 1 - 99,286\% = 0,714\%$$

$$(17) \mu = 90 \text{ km} \quad \sigma = 20 \text{ km} \quad X \sim N(\mu, \sigma^2)$$

$$a) P(X \geq 130) = 1 - \Phi\left(\frac{130 - 90}{20}\right) = 1 - \Phi(2) = 1 - 0,9772 = 2,28\%$$

$$b) 0,99 = 1 - \Phi\left(\frac{x - 90}{20}\right) \rightarrow \Phi^{-1}(0,99) = \frac{x - 90}{20} \rightarrow x = \underbrace{\Phi^{-1}(0,99)}_{= 2,33} \cdot 20 + 90 = 2,33 \cdot 20 + 90 = 46,6 + 90 = 136,6 \text{ km}$$

$$(18) \sigma = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{t^2}{2\sigma^2}} dt \quad ???$$

$$(19) Z = \frac{X - \mu}{\sigma} \rightarrow X = Z \cdot \sigma + \mu$$

$$Z \sim N(0,1)$$

$$X \sim N(\mu, \sigma^2)$$

$$\text{Beweis: } E[X] = E[Z\sigma + \mu] = \underbrace{E[Z]}_{0 \cdot \sigma} \cdot \sigma + \mu = \mu$$

$$\text{Var}[X] = \text{Var}[Z\sigma + \mu] = E[(Z\sigma + \mu - \mu)^2] = E[Z^2] \cdot \sigma^2 = \text{Var}[Z] \cdot \sigma^2 = \sigma^2$$

$$\text{Var}[X] = 1 \quad E[X] = ?$$

(20) gedächtnislos:

$$P(T > t+x | T > t) = \frac{P(A) \cap P(B)}{P(A)} = P(B) = \frac{P(A) \cdot P(B)}{P(A)} = P(B) \quad \checkmark$$

$$P(T > t+x) = P(A)$$

$$P(T > t)$$

$$\frac{P(T > t+x) \cdot P(T > t)}{P(T > t+x)} = P(T > t)$$

$$(21) F_X(x) = 1 - e^{-(\lambda x)^\beta}$$

$$(1) 0,5 = 1 - e^{-(\lambda \cdot 20)^\beta}$$

$$(2) 0,4 = 1 - e^{-(\lambda \cdot 40)^\beta}$$

$$(I) -0,5 = -e^{-(\lambda \cdot 20)^\beta}$$

$$\ln(0,5) = -\lambda \cdot 20^\beta$$

$$\ln(\ln(0,5)) = -\beta(\ln(20\lambda))$$

$$(II) -0,4 = -e^{-(\lambda \cdot 40)^\beta}$$

$$\ln(0,4) = -(\lambda \cdot 40)^\beta$$

$$\ln(\ln(0,4)) = -\beta(\ln(40\lambda))$$

$$\frac{I}{II} = \frac{\ln(\ln(0,5))}{\ln(\ln(0,4))} = \frac{\beta(\ln(\frac{20\lambda}{40\lambda}))}{\beta(\ln(\frac{20\lambda}{40\lambda}))} = \beta \cdot \ln(\frac{20\lambda}{40\lambda})$$

$$= \ln\left(\frac{-0,69315}{-0,91629}\right) = \beta \left(\ln\left(\frac{1}{2}\right)\right) \rightarrow \beta \ln(0,5)$$

$$= \frac{\ln\left(\frac{0,69315}{0,91629}\right)}{\ln(0,5)} = \beta = 0,403$$

$$(I) \ln(0,5) = -\lambda \cdot 20^{0,403}$$

$$\frac{\ln(0,5)}{-20^{0,403}} = 0,0207 = \lambda$$

Weibull Verteilung:

$$F_X(x) = 1 - e^{-(\lambda x)^\beta}$$

$$f_X(x) = \frac{d}{dx} (1 - e^{-(\lambda x)^\beta})$$

$$f_X(x) = -\beta(\lambda x)^{\beta-1} \cdot (-\lambda) \cdot e^{-(\lambda x)^\beta}$$

$$= \beta \lambda (\lambda x)^{\beta-1} \cdot e^{-(\lambda x)^\beta}$$

$$E[X] = \frac{1}{\lambda} \Gamma\left(1 + \frac{1}{\beta}\right)$$

$$\text{Var}[X] = \frac{1}{\lambda^2}$$

$$(22) \text{MTTF} = \frac{1}{\lambda} = 30000 \quad \lambda =$$

$$P(L) = P(L_1) \cap P(L_2) \cap P(L_3)$$

$$F_L(t) = 1 - e^{-\lambda t} \cdot P$$

$$F_{L_{ges}}(t) = (1 - e^{-\lambda t})^3$$

$$f_t(t) = \frac{d}{dt} (1 - e^{-\lambda t})^3 = 3(1 - e^{-\lambda t}) \cdot (-\lambda) \cdot e^{-\lambda t} = -3(1 - e^{-\lambda t})^2 \cdot \lambda e^{-\lambda t}$$

$$E[t] = \int_{-\infty}^{\infty} t \cdot f_t(t) dt \Rightarrow \int_{-\infty}^0 dt + \int_0^{\infty} -3\lambda e^{-\lambda t} (1 - e^{-\lambda t})^2 dt$$

$$= \underbrace{\int_0^{\infty} -3\lambda e^{-\lambda t} dt}_{\text{I}} + \underbrace{\int_0^{\infty} 6\lambda e^{-2\lambda t} dt}_{\text{II}} - \underbrace{\int_0^{\infty} 3\lambda e^{-3\lambda t} dt}_{\text{III}}$$

$$\text{I} \quad \int_0^{\infty} -3\lambda e^{-\lambda t} dt \quad \begin{matrix} u=t & u'=1 & v=e^{-\lambda t} & v'=-\lambda e^{-\lambda t} \end{matrix}$$

$$-3\lambda \left[-\frac{1}{\lambda} e^{-\lambda t} - \int +\frac{1}{\lambda} e^{-\lambda t} dt \right] = -3\lambda \left[-\frac{1}{\lambda} e^{-\lambda t} - \frac{1}{\lambda^2} e^{-\lambda t} \right]_0^{\infty} = +3\lambda \left[\frac{1}{\lambda} - \frac{1}{\lambda^2} \right] = 3t$$

$$\text{II} \quad \int_0^{\infty} 6\lambda e^{-2\lambda t} dt \Rightarrow \begin{matrix} v'=2\lambda e^{-2\lambda t} & v=-\frac{1}{2\lambda} e^{-2\lambda t} & u=t & u'=1 \end{matrix}$$

$$6\lambda \left[-\frac{1}{2\lambda} e^{-2\lambda t} - \int -\frac{1}{2\lambda} e^{-2\lambda t} dt \right]_0^{\infty} = 6\lambda \left[-\frac{1}{2\lambda} e^{-2\lambda t} + \frac{1}{4\lambda^2} e^{-2\lambda t} \right]_0^{\infty} = 6\lambda \left(\frac{1}{4\lambda^2} \right) = \frac{3}{2\lambda}$$

$$\text{III} \quad - \int_0^{\infty} 3\lambda e^{-3\lambda t} dt = +3\lambda \left[+\frac{1}{3\lambda} e^{-3\lambda t} - \int -\frac{1}{3\lambda} e^{-3\lambda t} dt \right]_0^{\infty} = -3\lambda \left[-\frac{1}{3\lambda} e^{-3\lambda t} - \frac{1}{9\lambda^2} e^{-3\lambda t} \right]_0^{\infty}$$

$$\text{I} \quad = +3\lambda \left[\frac{1}{\lambda} - \frac{1}{\lambda^2} \right] = -\frac{1}{3\lambda} \quad \text{II} \quad = 6\lambda \left[-\frac{1}{2\lambda} e^{-2\lambda t} + \frac{1}{4\lambda^2} e^{-2\lambda t} \right]_0^{\infty} = 3\lambda \left[\frac{1}{\lambda} - \frac{1}{\lambda^2} \right]_0^{\infty}$$

$$\text{I} + \text{II} + \text{III} = -\frac{3}{\lambda} + \frac{3}{2\lambda} - \frac{1}{3\lambda} = \frac{-18 + 9 - 2}{6\lambda} = -\frac{11}{6\lambda} = \frac{11}{6\lambda} = 55000$$

24) $N=10, n=4, R=4, r=2, G=6, g=2$

$$\frac{\binom{G}{g} \cdot \binom{R}{r}}{\binom{N}{n}} \rightarrow \frac{\binom{4}{2} \cdot \binom{6}{2}}{\binom{10}{4}} = \frac{\binom{4}{1} \cdot \binom{6}{3}}{\binom{10}{4}} + \frac{\binom{4}{0} \cdot \binom{6}{4}}{\binom{10}{4}} = \frac{30}{210} = \frac{1}{7} = 14,2\%$$

$n=4, r=2, R=4, R+G=10, G=6$

$$\frac{\binom{R}{r} \cdot \binom{G}{g}}{\binom{R+G}{n}} = \frac{\binom{4}{2} \cdot \binom{6}{2}}{\binom{10}{4}} + \frac{\binom{4}{1} \cdot \binom{6}{3}}{\binom{10}{4}} + \frac{\binom{4}{0} \cdot \binom{6}{4}}{\binom{10}{4}} = \frac{24 \cdot 3 \cdot 6 \cdot 5}{10 \cdot 9 \cdot 8 \cdot 7} = \frac{3}{7} + \frac{8}{21} + \frac{1}{14} = \frac{18}{42} + \frac{16}{42} + \frac{3}{42} = \frac{37}{42} = 88,1\%$$

$$\frac{\binom{4}{2} \cdot \binom{6}{2}}{\binom{10}{4}} = \frac{\frac{4 \cdot 3}{1 \cdot 2} \cdot \frac{6 \cdot 5}{1 \cdot 2}}{\frac{10 \cdot 9 \cdot 8 \cdot 7}{1 \cdot 2 \cdot 3 \cdot 4}} = \frac{6 \cdot 15}{210} = \frac{90}{210} = \frac{9}{21} = \frac{3}{7}$$

$$\frac{\binom{4}{1} \cdot \binom{6}{3}}{\binom{10}{4}} = \frac{4 \cdot \frac{6 \cdot 5 \cdot 4}{1 \cdot 2 \cdot 3}}{210} = \frac{80}{210} = \frac{8}{21}$$

$$\frac{\binom{4}{0} \cdot \binom{6}{4}}{\binom{10}{4}} = \frac{1 \cdot \frac{6 \cdot 5 \cdot 4 \cdot 3}{1 \cdot 2 \cdot 3 \cdot 4}}{210} = \frac{15}{210} = \frac{5}{70} = \frac{1}{14}$$

25) $\alpha = 3, \lambda = 12 \text{ s} = \mu \rightarrow \frac{\alpha}{\lambda} = 3:12, E[T] = T_1 + T_2 + T_3 = 12 \cdot 3 = 36$

$\downarrow \lambda = \frac{1}{12}$

$$P(T_1 + T_2 + T_3 \leq 60 \text{ s}) =$$

Verwendung der Gammafunktion: $\frac{\gamma(\alpha, \lambda \cdot x \cdot \lambda)}{(\alpha-1)!} = \frac{\gamma(3, 0,1 \cdot \frac{1}{12} \cdot 60)}{(3-1)!} = 0,875 \rightarrow 87,5\%$

26) Gamma-Funktion: $\frac{20}{60}$

$$F(t) = \frac{\gamma(3, 0,1 \cdot \frac{1}{12} \cdot 60)}{\Gamma(3)}$$

$$\Gamma(\alpha) = (\alpha-1)! = 2 \quad (\text{wenn } \alpha \in \mathbb{N} \rightarrow \Gamma(\alpha) = (\alpha-1)!)$$

$$F(t) = \frac{1,75}{2} = 0,875 = 87,5\%$$

$$F(t) = \frac{\gamma(\alpha, \beta, \lambda \cdot x)}{\Gamma(\alpha)} \quad (\beta \text{ in der VL nicht behandelt} \rightarrow \text{nur})$$

(27) Poisson-Verteilung: $p_k = \frac{\lambda^k}{k!} e^{-\lambda}$

$\frac{120}{360} \cdot 15 = \mu = 5 = \lambda \hat{=}$ mittlere Auftretenshäufigkeit des Ereignisses im betrachteten Zeitintervall

$P(X > 6) = 1 - P(X \leq 6) = 1 - \left(e^{-\lambda} \sum_{m=0}^6 \frac{\lambda^m}{m!} \right)$ $X \dots$ Anzahl Mitarbeiter $\lambda = \frac{120}{60} \cdot 15 = 5$

$P(X > 6) = 1 - e^{-5} \cdot \sum_{m=0}^6 \frac{5^m}{m!} = 23,78\%$

$\lambda = E[X] =$ Erwartungswert wie viele Anfragen bearbeitet werden

→ Poisson-Verteilung: in einem festen Zeitraum T genau k Ereignisse zu erhalten

(28) relativ seltenes Ereignis: $1 \text{ km}^2 \rightarrow 1 \text{ Blitzeinschlag}$

$\mu = \frac{20}{20} = 1 \cdot 5 = 5$

$P(X > 10) = 1 - P(X \leq 10) \quad k = X$

$= 1 - e^{-5} \cdot \sum_{k=0}^{10} \frac{5^k}{k!} = 1 - e^{-5} \cdot \left(\frac{5^0}{0!} + \frac{5^1}{1!} + \dots + \frac{5^{10}}{10!} \right) = 13,77\%$

(29) $\mu = \lambda = \frac{5}{10} = 0,5$

$P(X > 1) = 1 - P(X \leq 1) = 1 - e^{-0,5} \cdot \left(\frac{0,5^0}{0!} + \frac{0,5^1}{1!} \right) = 1 - e^{-0,5} (1 + 0,5) = 0,10902 = 10,902\%$

(30) $2,85 \Rightarrow E[X] = \frac{1}{n} \sum_{k=1}^n x_k$

$\text{Var}[X] = m_2 - \mu^2 = \frac{1}{n} \sum_{k=1}^n x_k^2 - \left(\frac{1}{n} \sum_{k=1}^n x_k \right)^2$

(31) $f(x, y) = f_x(x) \cdot f_y(y)$

$f(x|y) = \frac{f(x, y)}{f(y)} = f(x)$

$f(x, y) = f(x) \cdot f(y)$

$\frac{f(x, y)}{f(y)} = f(x) = f(x|y) \quad \text{bzw.} \quad \frac{f(x, y)}{f(x)} = f(y) = f(y|x)$

$\left. \begin{array}{l} f(x) = f(x|y) \\ f(y) = f(y|x) \end{array} \right\} \quad \frac{f(x)}{f(y)} = \frac{f(x|y)}{f(y)} = \frac{f(y)}{f(y|x)} = \frac{f(y)}{f(y)}$

$$(33) \rho[X,Y] := \frac{\text{Cov}[X,Y]}{\sqrt{\text{Var}[X] \cdot \text{Var}[Y]}}$$

$$\begin{aligned}\text{Cov}[X,Y] &= E[(X-\mu_X) \cdot ((aX+b)-\mu_Y)] \\ &= E[(X-\mu_X) \cdot (aX+b - E[aX+b])] \\ &= E[(X-\mu_X) \cdot (aX+b - a\mu_X - b)] \\ &= E[(X-\mu_X) \cdot (a(X-\mu_X))] \\ &= E[a(X-\mu_X)^2] = a \cdot E[(X-\mu_X)^2] \\ &= E[(X-\mu_X)^2] \cdot a = \text{Var}[X] \cdot a\end{aligned}$$

$$\hookrightarrow \text{Var}[Y] = \text{Var}[aX+b] = \text{Var}[X] \cdot a^2$$

$$\rho[X,Y] = \frac{\text{Cov}[X,Y]}{\sqrt{\text{Var}[X] \cdot \text{Var}[X] \cdot a^2}} = \frac{a \cdot \text{Var}[X]}{\pm \text{Var}[X] \cdot a} = \frac{a}{\pm a} = \text{sign}(a)$$

Der Korrelationskoeffizient liegt zwischen -1 und 1, je größer sein Absolutwert, desto stärker ist die lineare Beziehung zwischen den Variablen. $\rho = 0 \rightarrow$ keine lineare Beziehung.

$$(34) f(a,b) = E[(Y - (aX+b))^2] = E[(Y^2 - 2Y(aX+b) + (aX+b)^2)]$$

$$f(a,b) = E[(Y-\mu_Y - a(X-\mu_X) + (\mu_Y - (a\mu_X+b)))^2]$$

$$E[(Y-\mu_Y)^2 + a^2(X-\mu_X)^2 - 2a(Y-\mu_Y)(X-\mu_X) + 2(Y-\mu_Y)(\mu_Y - (a\mu_X+b)) - a(X-\mu_X)(\mu_Y - (a\mu_X+b))]$$

$$E[Y^2] + a^2 E[X^2] - 2a \cdot \text{Cov}[X,Y] + E[(\mu_Y - (a\mu_X+b))^2] + 2E[(Y-\mu_Y)(\mu_Y - (a\mu_X+b))] - 2a E[(X-\mu_X)(\mu_Y - (a\mu_X+b))] \quad *$$

$$\text{NR: } 2E[(Y-\mu_Y)(\mu_Y - (a\mu_X+b))] = 2E[\mu_Y^2 - \mu_Y^2 - \mu_Y(a\mu_X+b) + \mu_Y(a\mu_X+b)] = 0, \quad (\mu_X(a\mu_X+b)) = 0$$

$$2a E[(X-\mu_X)(\mu_Y - (a\mu_X+b))] = 2a(\mu_X\mu_Y - \mu_X\mu_Y - \mu_X(a\mu_X+b) + \mu_X(a\mu_X+b)) = 0$$

$$E[\mu_Y - (a\mu_X+b)^2] = E[\mu_Y - (a^2\mu_X^2 + 2ab\mu_X + b^2)] = \mu_Y - a^2\mu_X^2$$

$$* \quad E[Y^2] + a^2 E[X^2] - 2a \cdot \text{Cov}[X,Y] + (\mu_Y - (a\mu_X+b))^2 = f(a,b) = \min$$

$$df/da = 2a E[X^2] - 2 \text{Cov}[X,Y] - 2(\mu_Y - a\mu_X - b)(\mu_X) \stackrel{!}{=} 0 \quad \text{I}$$

$$df/db = 2(\mu_Y - a\mu_X - b)(-1) \stackrel{!}{=} 0 \quad \text{II}$$

$$2a E[X^2] - 2\mu_X \mu_X \cdot (\mu_Y - \mu_Y) - 2a\mu_X - 2a\mu_X$$

$$\textcircled{I} \quad -\mu_y + a\mu_x + b = 0 \rightarrow b = -a\mu_x + \mu_y \quad a\mu_x - \mu_y = 0$$

$$\textcircled{I} \quad 2a\sigma_x^2 - 2\text{cov}(X,Y) - 2\mu_x(\mu_y - (a\mu_x + \mu_y - a\mu_x)) = 0$$

$$\rightarrow a\sigma_x^2 - \text{cov}(X,Y) - \mu_x(\mu_y - a\mu_x - \mu_y + a\mu_x)$$

$$a = \frac{\text{cov}(X,Y) - \mu_x\mu_y - \mu_x^2}{\sigma_x^2}$$

$\textcircled{35}$

35) 2 Sensoren $\rightarrow$ 2 stochastisch unabhängige Exponentialverteilung

$$\left. \begin{array}{l} 1. \text{ Sensor: } \lambda e^{-\lambda x} \\ 2. \text{ Sensor: } \lambda e^{-\lambda x} \end{array} \right\} x \geq 0 \quad F(t) = 1 - e^{-\lambda t}$$

a) Erwartungswert: kann addiert werden! $E[X] = E[X_1] + E[X_2] = \frac{1}{\lambda} + \frac{1}{\lambda} = \frac{2}{\lambda}$

b) Dichtefunktion:

$$f_T(t) = \int_{-\infty}^{\infty} f(x) \cdot f(t-x) dx \quad \text{Grenzen } t-x \geq 0 \quad x \geq 0 \quad \rightarrow \quad t \leq x \quad 0 \leq x \leq t$$

$$\begin{aligned} f_T(t) &= \int_0^t f(x) \cdot f(t-x) dx = \int_0^t \lambda e^{-\lambda x} \cdot \lambda e^{-\lambda(t-x)} dx \\ &= \lambda^2 \int_0^t e^{-\lambda x - \lambda(t-x)} dx = \lambda^2 \int_0^t e^{-\lambda(t)} dx = \left[\lambda^2 x e^{-\lambda t} \right]_0^t = \underline{\underline{\lambda^2 t e^{-\lambda t}}} \end{aligned}$$

$$f_T(t) = \begin{cases} \lambda^2 t e^{-\lambda t}, & t \geq 0 \\ 0, & t \leq 0 \end{cases}$$

36) $N(\mu_i, \sigma_i^2)$ $\mu_1 = 3$ $\mu_2 = 2$ $\sigma_1 = 4$ $\sigma_2 = 3$ $\text{Var}[X_1] = \sigma_1^2$ $\text{Var}[X_2] = \sigma_2^2$

$$E[X_1] + E[X_2] = 3 + 2 = 5 = E[X]$$

$$\text{Var}[X] = \text{Var}[X_1] + \text{Var}[X_2] = 4^2 + 3^2 = 25 \rightarrow \sigma = 5$$

$$X+Y \sim N(5, 25)$$

$$P(X \leq 10) = \Phi\left(\frac{10-5}{5}\right) = \Phi(1) = 84,13\%$$

37) $E[X] = \int_{x=-\infty}^{x=\infty} \frac{1}{b-a} dx = \int_{x=-\infty}^{x=a} 0 dx + \int_{x=a}^{x=b} \frac{x}{b-a} dx + \int_{x=b}^{x=\infty} 0 dx = \left[\frac{1}{2} \frac{x^2}{b-a} \right]_a^b = \frac{b^2 - a^2}{2(b-a)} = \frac{b+a}{2}$

$$E[Z] \stackrel{!}{=} 0 \rightarrow \sum_{i=1}^k \frac{1}{2} \rightarrow \frac{k}{2} = b$$

per Definition $\begin{cases} E[U] = \frac{1}{2} \\ \text{Var}[U] = \frac{1}{12} \end{cases} \quad \begin{matrix} b=1 \\ a=0 \end{matrix}$

$$\text{Var}[Z] \stackrel{!}{=} 1 \rightarrow \text{Var}\left[\sum_{i=1}^k U_i - b\right] = 1$$

$$1 = \text{Var}\left[\sum_{i=1}^k U_i\right] = \frac{k \cdot (b-a)^2}{12} \Rightarrow 1 \rightarrow k = 12 \rightarrow b = 6$$

38) a) $\mu = 5s$ $\sigma = 10s$

$$\mu_s = 121 \cdot 5s \quad \sigma_s^2 = 11 \cdot \sigma^2 = 121 \cdot 100 = 12100$$

$$P(X \leq 660s) = \Phi\left(\frac{660 - 605}{\sqrt{12100}}\right) = \Phi(0,5) = 69,15\%$$

b) $0,99 = \Phi\left(\frac{X - 605}{110}\right)$

$$\Phi^{-1}(0,99) = 110 + 605 = X$$

$$X = 214 \cdot 110 + 605 = 869s = 14 \text{ min } 29s$$